

COE428 – VLSI Design  
Course Project  
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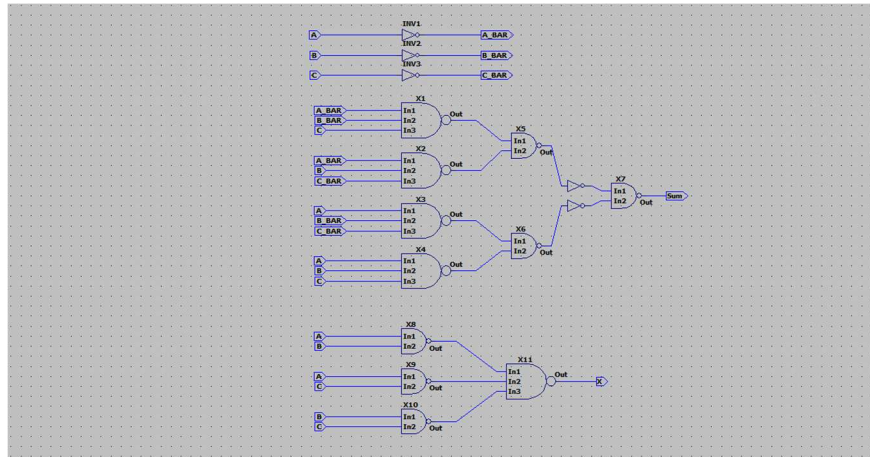
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# Part 1: Schematic

## Top Level Circuit and Transistor Sizing

With no design requirements beyond transistor rise and fall resistances, little effort was made beyond achieving a steady-state output equivalent to that of the full-adder equation. Our design is as follows (from *Full\_Adder.asc* file):

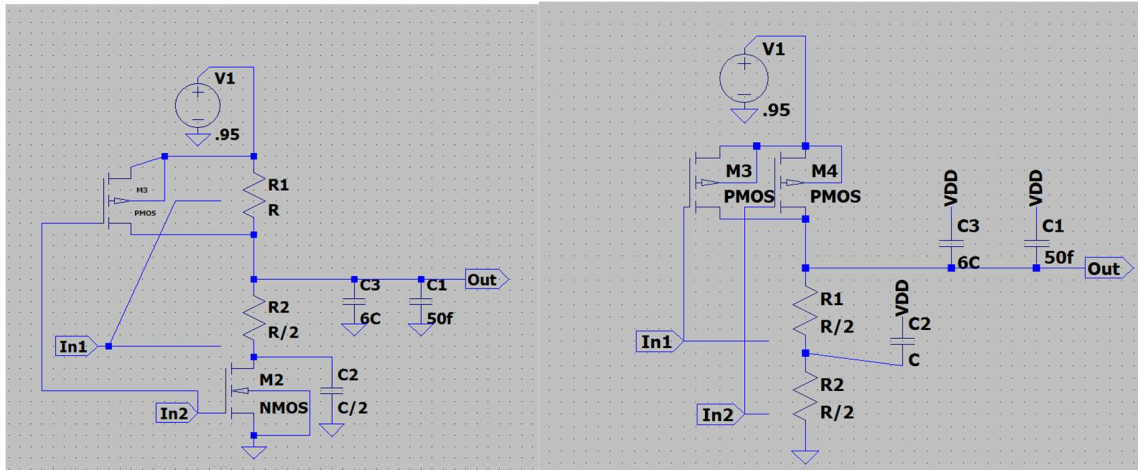
INV(P:N = 2:1), NAND2(P:N = 2:2), NAND3(2:3)



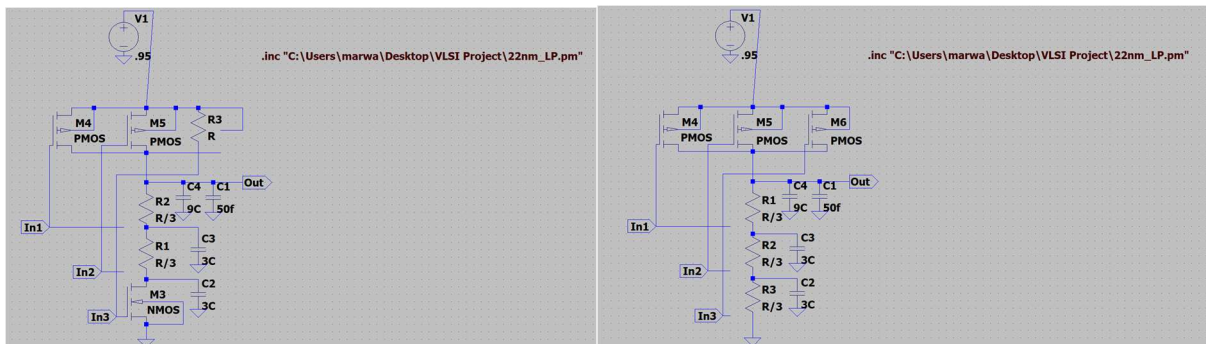
## Timing Delay

The selection paths for the worst propagation delays are as follows (figures are illustrative and only show to serve resistor and capacitor locations):

NAND2:



NAND3:



To measure propagation delays, a method using SPICE derivatives was employed. Specifically, I used “*.meas TRAN*” which accepts arguments relating to trigger values over the generated time-series data. Besides this, I also verified using the cursor method later; of course, the SPICE derivative method is more accurate as it uses the solver’s engine to interpolate the result at a much lower error tolerance value than the cursor. The general structure is as follows (all commands can be found in the *CKT\_Test.asc* schematic file):

For rising delay (triggered by a rising input):

```
.meas TRAN tpdf_gate_x TRIG V(in) VAL=0.475 RISE=1  
+ TARG V(out) VAL= 0.475 FALL=1
```

For rising delay (triggered by a rising input):

```
.meas TRAN tpdr_gate_x TRIG V(in) VAL=0.475 RISE=1  
+ TARG V(out) VAL= 0.475 FALL=1
```

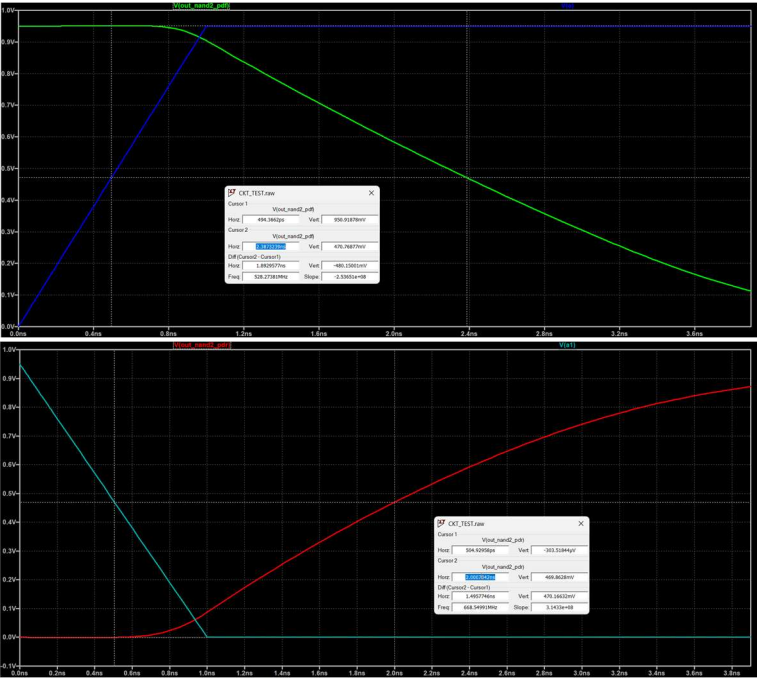
Thus, the values were accessed from the log file generated by running *CKT\_Test.asc* at the bottom (or upon pressing CTRL+L in LTspice):

```
tpdf_inv=1.80722e-09 FROM 5e-10 TO 2.30722e-09  
tpdr_inv=1.5138e-09 FROM 4.15e-08 TO 4.30138e-08
```

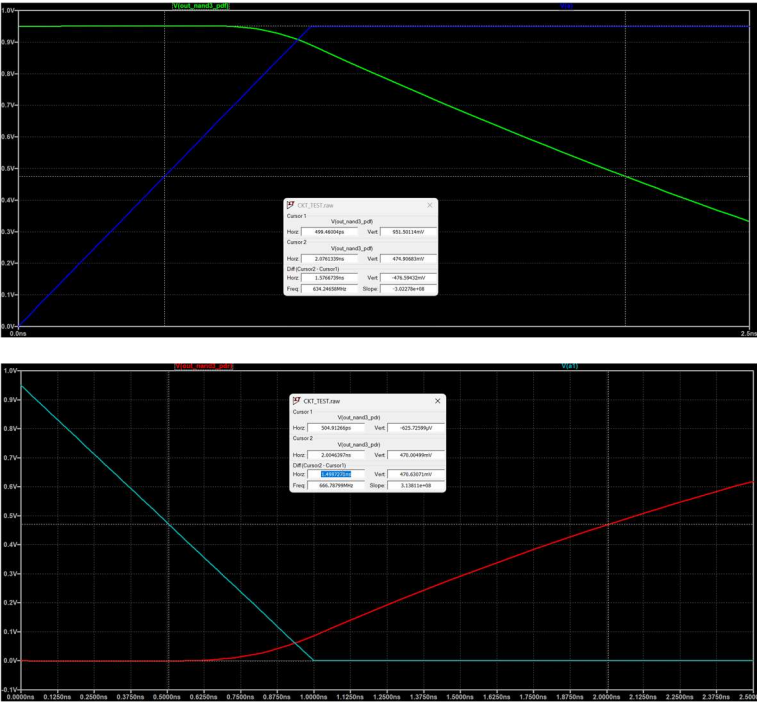
```
tpdf_nand2=1.87135e-09 FROM 5e-10 TO 2.37135e-09  
tpdr_nand2=1.51637e-09 FROM 5e-10 TO 2.01637e-09
```

```
tpdf_nand3=1.94962e-09 FROM 5e-10 TO 2.44962e-09  
tpdr_nand3=1.51999e-09 FROM 5e-10 TO 2.01999e-09
```

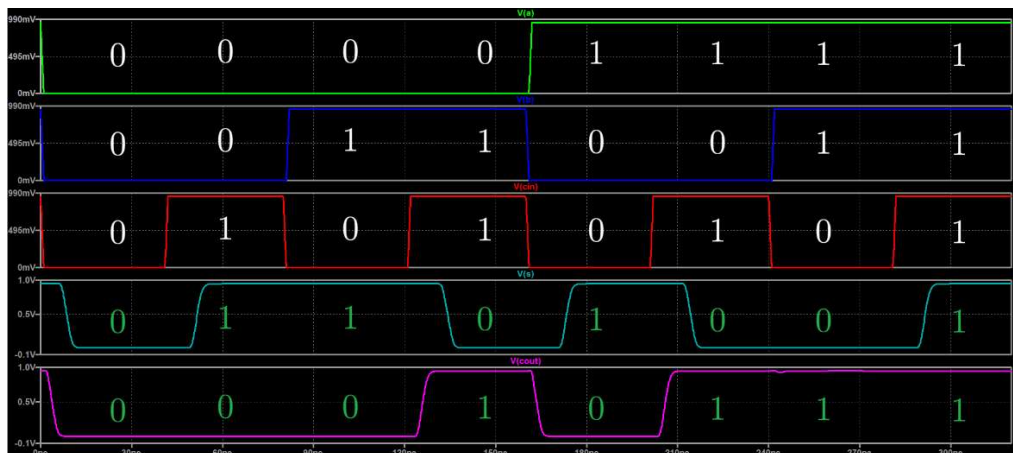
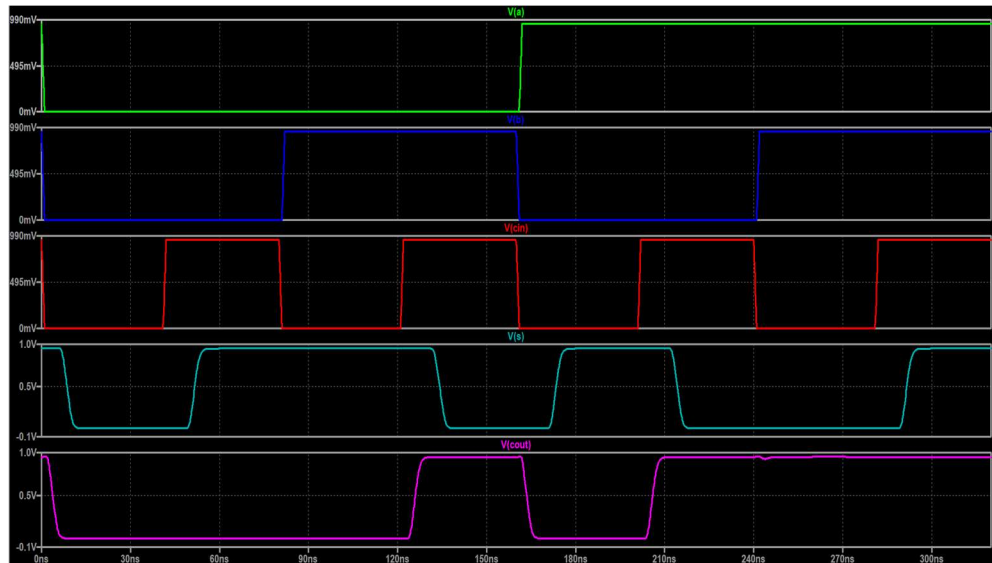
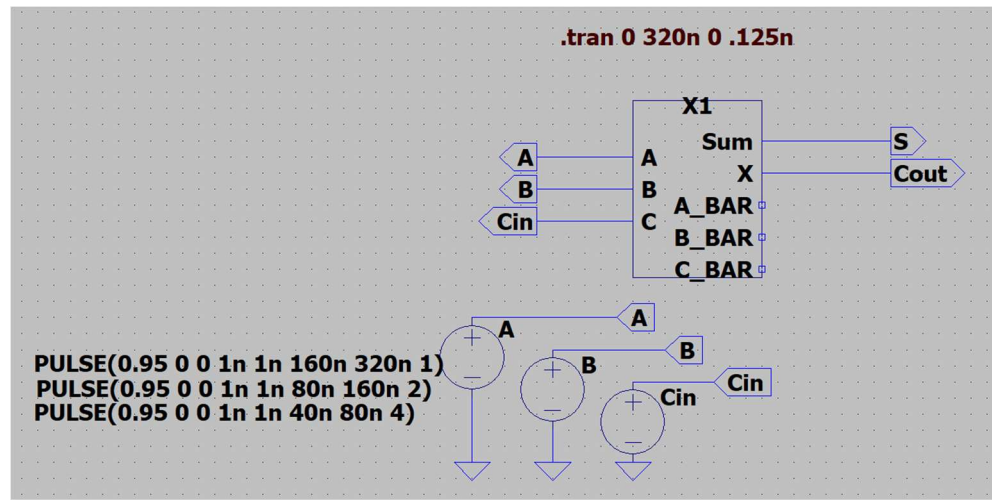
NAND2:



NAND3:



The top module schematic of the full adder can be found in the file *Circuit.asc*. Below are the simulation results for a 320ns transient simulation time, solver settings are all defaulted:



## Top level Module Delay

```
tpdr1_fulladder_sum=1.00776e-08 FROM 4.15e-08 TO 5.15776e-08
tpdr2_fulladder_sum=1.28868e-08 FROM 1.605e-07 TO 1.73387e-07
tpdr3_fulladder_sum=1.00826e-08 FROM 2.815e-07 TO 2.91583e-07
tpdf1_fulladder_sum=8.17253e-09 FROM 5e-10 TO 8.67253e-09
tpdf2_fulladder_sum=1.28095e-08 FROM 1.215e-07 TO 1.3431e-07
tpdf3_fulladder_sum=1.2838e-08 FROM 2.015e-07 TO 2.14338e-07
tpdr1_fulladder_cout=4.40524e-09 FROM 1.215e-07 TO 1.25905e-07
tpdr2_fulladder_cout=4.39998e-09 FROM 2.015e-07 TO 2.059e-07
tpdf1_fulladder_cout=3.3852e-09 FROM 5e-10 TO 3.8852e-09
tpdf2_fulladder_cout=3.10332e-09 FROM 1.605e-07 TO 1.63603e-07
```

Furthermore, the worst-case scenario is as follows, where the sequence in paranthesis represents  $(ABC_{old} \rightarrow ABC_{new})$  respectively:

Sum (path starting from input C):

| Rising propagation delay (011 $\rightarrow$ 100) | Falling propagation delay (100 $\rightarrow$ 101) |
|--------------------------------------------------|---------------------------------------------------|
| 12.88ns                                          | 12.84ns                                           |

Cout (path starting from input C):

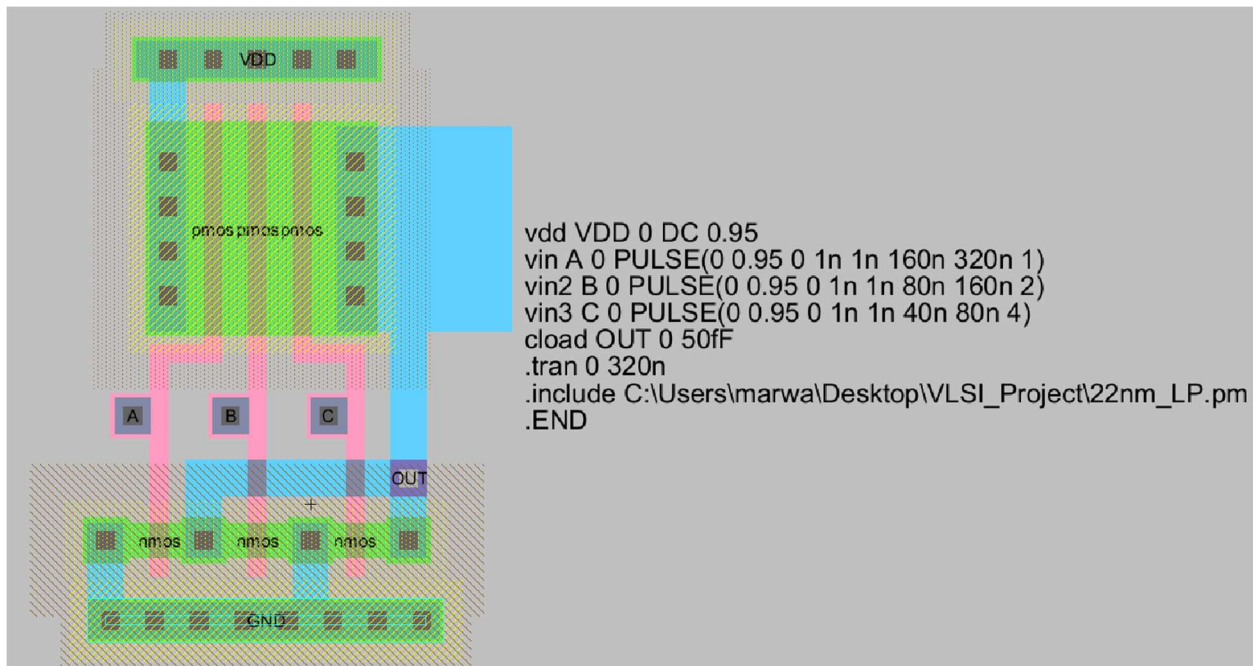
| Rising propagation delay (010 $\rightarrow$ 011) | Falling propagation delay (111 $\rightarrow$ 000) |
|--------------------------------------------------|---------------------------------------------------|
| 4.405ns                                          | 3.38ns                                            |

These values align to the worst paths considered from the individual gates from the previous subsection.



## Part 2: Layout

### Electric VLSI Design Cell Layout



```
Running DRC with area bit on, extension bit on, Mosis bit
Checking again hierarchy .... (0.009 secs)
Found 15 networks
Checking cell 'NOR3{lay}'
    No errors/warnings found
0 errors and 0 warnings found (took 0.212 secs)
=====7=====
Checking Wells and Substrates in 'VLSI_DESIGN_PROJ_P2:NOR3{lay}' ...
    Geometry collection found 24 well pieces, took 0.012 secs
    Geometry analysis used 16 threads and took 0.007 secs
NetValues propagation took 0.0 secs
Checking short circuits in 2 well contacts
    Additional analysis took 0.008 secs
No Well errors found (took 0.03 secs)
```

## Generated Netlist:

```
*** TOP LEVEL CELL: NOR3{lay}

Mnmos@0 OUT A gnd gnd nmos L=0.4U W=0.8U AS=3.68P AD=2.56P PS=9.533U PD=5.8U
Mnmos@7 gnd B OUT gnd nmos L=0.4U W=0.8U AS=2.56P AD=3.68P PS=5.8U PD=9.533U
Mnmos@8 OUT C gnd gnd nmos L=0.4U W=0.8U AS=3.68P AD=2.56P PS=9.533U PD=5.8U
Mpmos@0 net@0 A vdd vdd pmos L=0.4U W=4.8U AS=11.84P AD=1.59P PS=25.4U PD=5.9U
Mpmos@1 net@1 B net@0 vdd pmos L=0.4U W=4.8U AS=1.59P AD=1.59P PS=5.9U PD=5.9U
Mpmos@3 OUT C net@1 vdd pmos L=0.4U W=4.8U AS=1.59P AD=2.56P PS=5.9U PD=5.8U

* Spice Code nodes in cell cell 'NOR3{lay}'

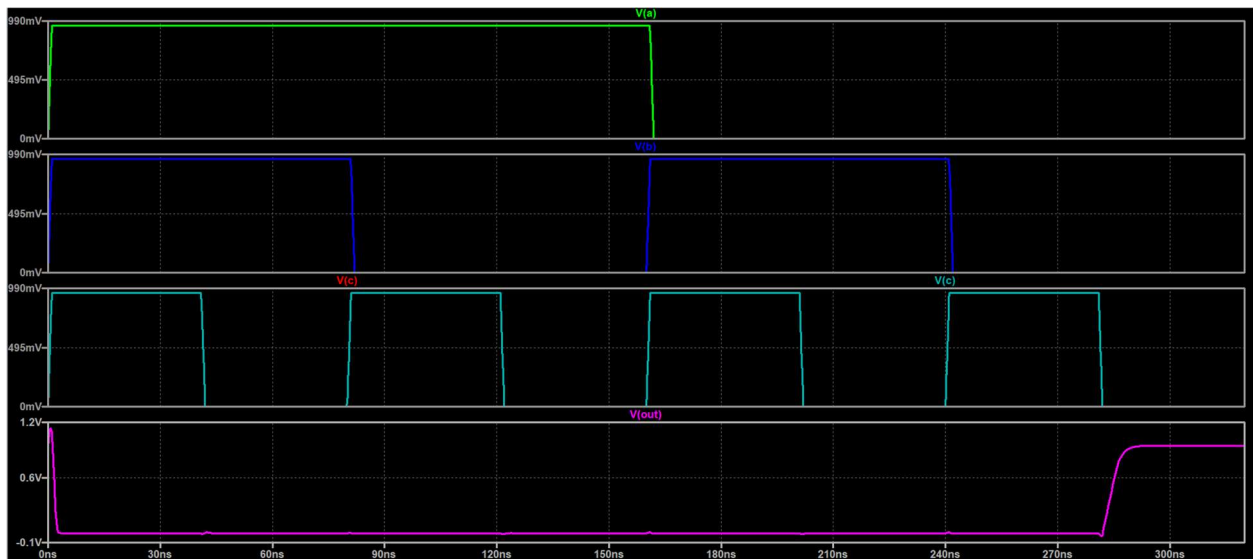
vdd VDD 0 DC 0.95

vin A 0 PULSE(0 0.95 0 1n 1n 160n 320n 1)
vin2 B 0 PULSE(0 0.95 0 1n 1n 80n 160n 2)
vin3 C 0 PULSE(0 0.95 0 1n 1n 40n 80n 4)

cload OUT 0 50fF

.tran 0 320n

.include C:\Users\marwa\Desktop\VLSI Project\22nm_LP.pm
```



## Timing Delay

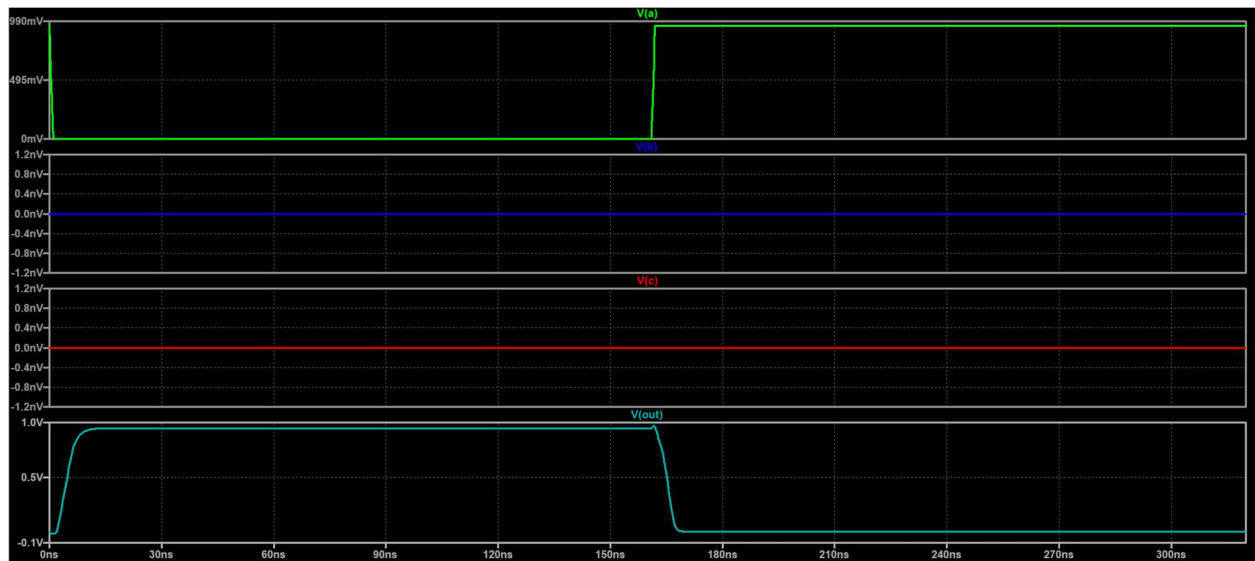
$T_{pdf}$ :

```
vdd VDD 0 DC 0.95
vin A 0 PULSE(0.95 0 0 1n 1n 160n 320n 1)
vin2 B 0 0
vin3 C 0 0
cload OUT 0 50fF
.tran 0 320n
.include C:\Users\marwa\Desktop\VLSI Project\22nm_LP.pm

.MEAS TRAN tpdf_nor3 TRIG V(a) VAL=0.475 RISE=1
+ TARG V(out) VAL= 0.475 FALL=1
.END
```

Output:

**tpdf\_nor3=3.7538e-09 FROM 1.615e-07 TO 1.65254e-07**

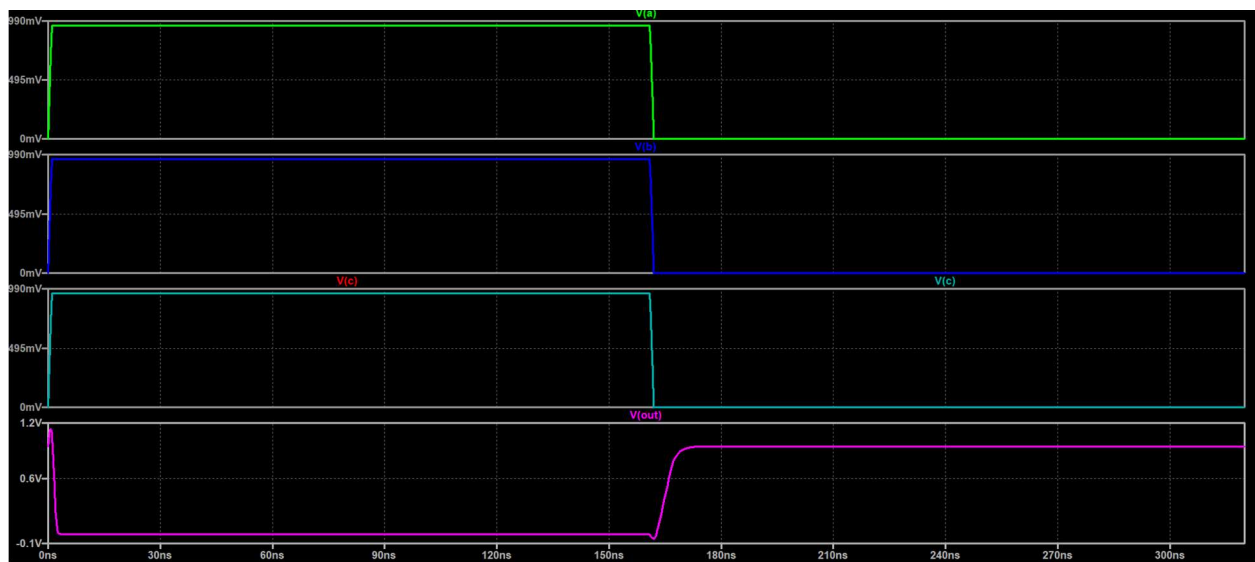


T<sub>pdr</sub>:

```
* Spice Code nodes in cell cell 'NOR3{lay}'  
vdd VDD 0 DC 0.95  
vin A 0 PULSE(0 0.95 0 1n 1n 160n 320n 1)  
vin2 B 0 PULSE(0 0.95 0 1n 1n 160n 320n 1)  
vin3 C 0 PULSE(0 0.95 0 1n 1n 160n 320n 1)  
cload OUT 0 50fF  
.tran 0 320n  
.include C:\Users\marwa\Desktop\VLSI Project\22nm_LP.pm  
  
.MEAS TRAN tpdr_nor3 TRIG V(a) VAL=0.475 FALL=1  
+ TARG V(out) VAL= 0.475 RISE=1  
.END
```

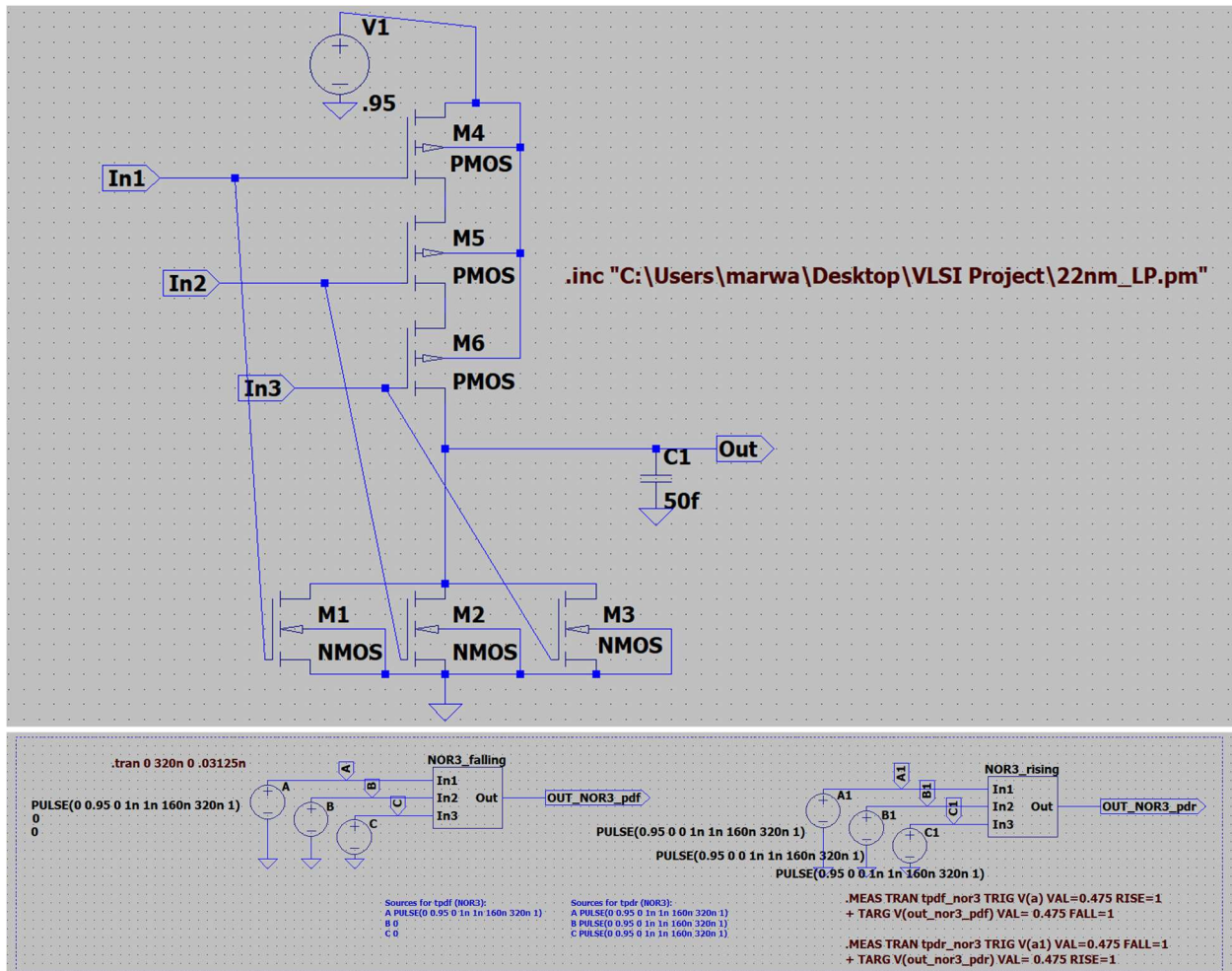
Output:

tpdr\_nor3=3.84793e-09 FROM 1.615e-07 TO 1.65348e-07



## LTspice native circuit simulation:

(P:N = 6:1) PMOS (W = 264nm; L = 22nm) NMOS (W = 44nm; L = 22nm)



## Timing Delay

Using the same method as in part a through the “.meas” SPICE directive to obtain propagation delays, The following comparison can be made:

| NOR3             | Electric  | LTspice   |
|------------------|-----------|-----------|
| T <sub>pdf</sub> | 3.84793ns | 1.84651ns |
| T <sub>pd</sub>  | 3.7538ns  | 2.06641ns |

Clearly, LTspice results align with an RC-dominated delay behavior, indicating that the Elmore delay model (or logical effort method) provides a reasonable first-order approximation. In contrast, Electric incorporates layout-dependent parasitic capacitances, which increases the overall delay and can introduce deviations from the ideal RC model. These discrepancies highlight the necessity of parasitic extraction and iterative layout refinement in a typical VLSI design workflow using Electronic Design Automation (EDA) tools.